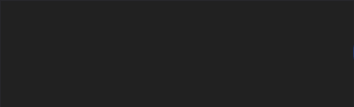


Project Details

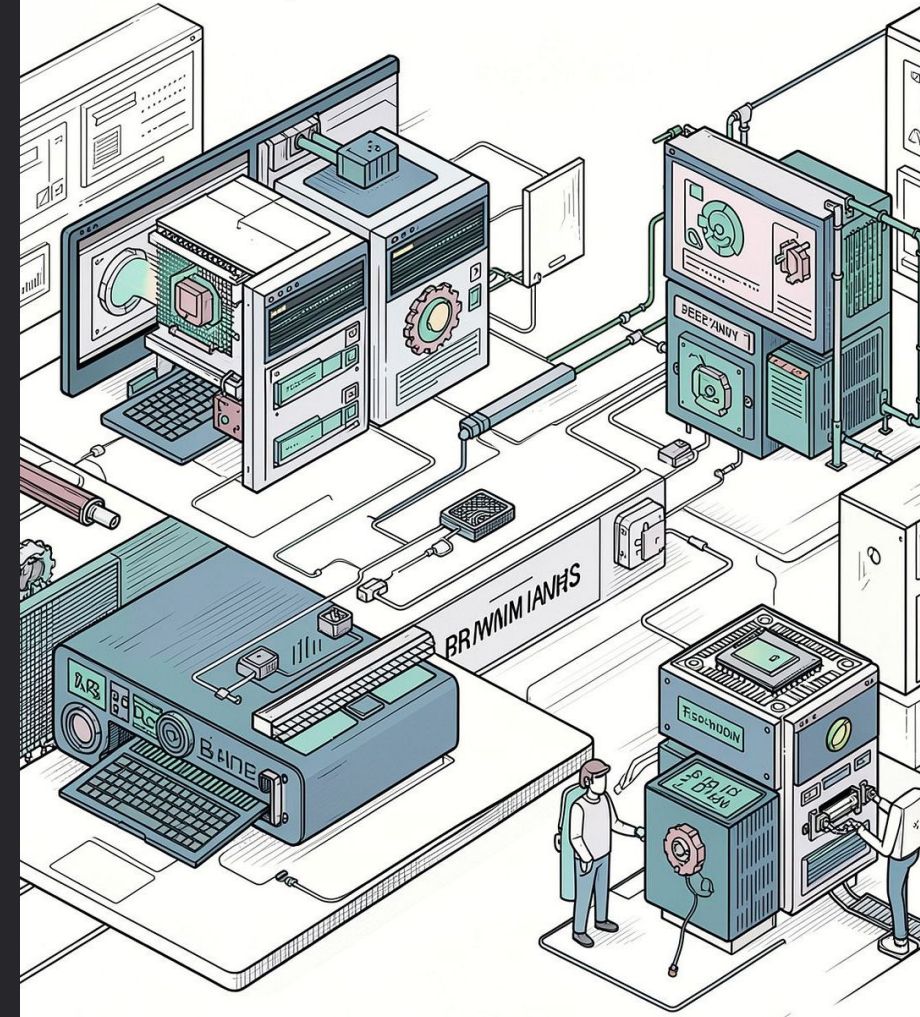
Field	Value	Field	Value
Date	30 September 2026	Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant		
Maximum Marks	1 Mark	Evaluator	—



Section: Scalability

EduGenie is engineered for modular expansion. The architecture separates core web services from AI components so new capabilities, higher concurrency, and improved models can be integrated with minimal disruption. The system supports staged scaling across backend, model, feature, user, and data layers.

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Scalability — Technical Areas

Category	Description
Backend Scalability	The FastAPI backend can be expanded using modular REST endpoints and asynchronous request handling to support increased user requests and additional services.
AI Model Scalability	The Gemini-powered architecture allows supported AI models to be updated or replaced as newer models become available, while maintaining the same application workflow.
Feature Scalability	New educational features such as flashcards, assignment assistance, adaptive assessments, voice interaction, and multilingual learning can be added as separate modules.
User Scalability	The application can be deployed on scalable cloud infrastructure to support a larger number of concurrent students.
Data Scalability	Future versions can introduce persistent databases and learning-history storage to manage student profiles, progress, quiz results, and personalized recommendations.

Section: Future Plan

Planned enhancements focus on accessibility, personalization, analytics, and deployment readiness. Each item is scoped as a modular extension to preserve the baseline application and allow phased rollout aligned with evaluation milestones.

Future Plan — Roadmap Items

#	Enhancement	Description
1	Voice-Based Learning	Allow students to interact with EduGenie using speech for questions, explanations, and learning assistance.
2	Multilingual Support	Provide educational explanations, summaries, and answers in multiple languages to make the system accessible to more learners.
3	Adaptive Learning	Use student performance, quiz results, and learning history to dynamically adjust recommendations and difficulty levels.
4	Personalized Student Profiles	Maintain individual learning preferences, progress, completed topics, and quiz performance.
5	Gamification	Introduce points, badges, progress tracking, challenges, and achievement-based learning activities.
6	Advanced Learning Analytics	Provide insights into student strengths, weaknesses, learning progress, and frequently requested topics.
7	Improved AI Accuracy	Continuously improve prompts, validation, model selection, and response processing to produce more reliable educational answers.
8	Cloud Deployment	Deploy the complete EduGenie application on scalable cloud infrastructure for wider student access.

Implementation Considerations

Phased Rollout

Deliver core features first, then add voice and multilingual modules in subsequent sprints.

Security & Privacy

Design data storage and analytics with student privacy, consent, and encryption best practices.

Monitoring & Ops

Instrument telemetry for performance, model quality, and user behavior to guide improvements.

Risk & Mitigation

Primary Risks

- Model hallucination or inaccurate responses
- Data privacy and compliance
- Scalability under peak load

Mitigations

- Prompt validation, response verification, and human-in-the-loop review
- Encryption, access controls, and minimal data retention
- Autoscaling infrastructure and load testing

Evaluation Metrics



Learning Outcomes

Measure concept mastery improvements via pre/post assessments and adaptive quizzes.



Engagement

Track active sessions, retention, and feature adoption (voice, flashcards, profiles).



Accuracy

Monitor model response correctness and reduce error rates through validation pipelines.

Conclusion & Contact

EduGenie is architected for modular growth, focusing on practical scalability and learner-centered enhancements. The roadmap emphasizes accessibility, personalization, and trustworthy AI usage while preserving the project's original structure for straightforward evaluation and deployment.

Contact: EduGenie Team — SWTID-2026-3580

